Skill, Sequence, and Scoring: A Mathematical Comparison of Traditional and World Bowling Scoring Systems

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Abstract

Ten-pin bowling's traditional scoring system rewards sustained performance: strikes and spares earn bonus points from subsequent balls, creating dependencies across frames. World Bowling, the international governing body, has promoted a simpler alternative in which each frame is scored independently. What exactly is lost when these dependencies are removed? Part 1 uses exact combinatorial enumeration to prove that current-frame scoring is mathematically invariant to the order of frames: two players who throw identical frames in different sequences always receive the same score, whereas traditional scoring produces a range of up to 39 points across orderings of identical frames. Part 2 uses Monte Carlo simulation across six skill tiers, calibrated against Professional Bowlers Association data, to identify what this invariance costs. A variance decomposition shows that the two systems are about equally reliable at distinguishing players who differ in raw skill: single-game reliability is nearly identical across the full skill range, and traditional scoring's larger spread at elite levels is mostly within-player noise. The cost of commutativity is specific. At a fixed strike rate, traditional scores rise with a player's tendency to string strikes together (correlation 0.49 with streakiness), whereas current-frame scores stay flat (correlation −0.04). Current-frame scoring does not measure skill less accurately; it is structurally blind to the one dimension of skill that sustained, consecutive striking represents. The complexity of traditional scoring is the mechanism by which it encodes that dimension.

1 Introduction

Ten-pin bowling has been scored using the same traditional system for over a century. Under this system, a strike earns ten pins plus the total of the next two balls thrown, and a spare earns ten pins plus the next ball. This forward-looking bonus structure creates scoring dependencies across frames, rewarding sustained performance non-linearly.

World Bowling, the international governing body for the sport, has promoted an alternative current-frame scoring system (World Bowling, 2014) in which each frame is scored independently: a strike earns 30 points, a spare earns 10 plus the first ball of that frame, and an open frame earns the sum of both balls. The stated motivation is simplicity: spectators and new participants can follow the score without tracking future balls. This push has intensified as the International Bowling Federation pursues Olympic inclusion, where broadcast accessibility is a key selection criterion (Inside The Games, 2014), and has culminated in the launch of international franchise competitions using current-frame scoring exclusively (Chakraborty, 2025). The question of whether this simplification preserves the competitive properties the sport requires at the elite level is therefore not academic. It is an active governance decision with immediate consequences.

A simple example illustrates the core difference. Consider two players who each throw five strikes and four spares across frames 1–9, followed by an identical frame 10. Under World Bowling, both players score 219, regardless of whether the strikes were consecutive or scattered throughout the game. Under traditional scoring, the player who placed strikes consecutively scores up to 20 points higher than one who alternated. The same frames, in different order, produce the same World Bowling score but meaningfully different traditional scores. This paper asks: does this distinction matter, and at what skill level does it become competitively significant?

Several informal comparisons of these systems exist online, focusing primarily on average scores and the presence of impossible scores in the World Bowling range. Cooper & Kennedy (1990) provided the foundational mathematical treatment of traditional bowling score distributions using generating functions, later extended by Hohn (2009) to generalised N-frame, M-pin games. Keogh & O'Neill (2011) studied empirical bowling score distributions using real game data. However, no prior work has formally compared the mathematical properties of both scoring systems, characterised sequence sensitivity as a distinguishing property, or quantified the skill level at which scoring system choice becomes competitively significant.

This paper provides that treatment in two complementary parts: exact mathematical analysis (Part 1) and skill-weighted simulation (Part 2).

Part 1: Mathematical Analysis

2 Mathematical Framework

2.1 State Space and Enumeration

The total number of distinct ten-pin bowling games is finite and well-defined. Under traditional scoring, each of frames 1–9 admits 66 possible throwing outcomes (one strike, plus all combinations of two balls summing to at most 10), and frame 10 admits 241 outcomes (accounting for bonus balls). The total game count is therefore:

$$66^9 \times 241 = 5{,}726{,}805{,}883{,}325{,}784{,}576 \approx 5.7 \times 10^{18}$$

Under World Bowling scoring, all ten frames are identical (no bonus ball in frame 10), giving:

$$66^{10} = 1{,}568{,}336{,}880{,}910{,}795{,}776 \approx 1.6 \times 10^{18}$$

Direct enumeration of these game spaces is computationally infeasible, and the resulting distributions are highly multimodal and non-normal, as demonstrated empirically by McCarthy (2011) using PBA tournament data, ruling out simple parametric approximations. We therefore employ dynamic programming with state representation tracking pending bonus multipliers, following the convolution approach described by Balmoral Software (2019). The key insight is that the score contribution of each ball depends only on the current pending bonus state, not on the full history of prior balls, allowing the computation to proceed frame by frame with a manageable number of distinct states (on the order of hundreds of thousands, rather than quintillions of complete games).

Our independent implementation produces results that match their published tables exactly at every verification point. These include the mode of 77 for traditional scoring with 172,542,309,343,731,946 possible games, and the mode of 72 for World Bowling scoring with 43,781,414,679,391,290 possible games.

2.2 Scoring Functions

Traditional scoring: For frames 1–9, let A_i and B_i denote the first and second balls of frame i. For a strike ($A_i = 10$), the frame score is $10 + A_{i+1} + B_{i+1}$. For a spare ($A_i + B_i = 10$), the frame score is $10 + A_{i+1}$. Otherwise the frame score is $A_i + B_i$. Frame 10 follows special rules permitting up to three balls with no bonus propagation beyond the frame.

World Bowling scoring: For each frame i, the score is: 30 (strike), $10 + A_i$ (spare), or $A_i + B_i$ (open). All ten frames are scored identically; no inter-frame dependencies exist.

3 Score Distributions

3.1 Summary Statistics

Table 1: Summary statistics for exact score distributions under uniform random play.

Statistic	Traditional	World Bowling
Total games	5.73×10^{18}	1.57×10^{18}
Score range	0–300	0–289, 300
Mode	77	72
Mean	79.7	76.5
Median	79	75
Standard deviation	13.7	15.2

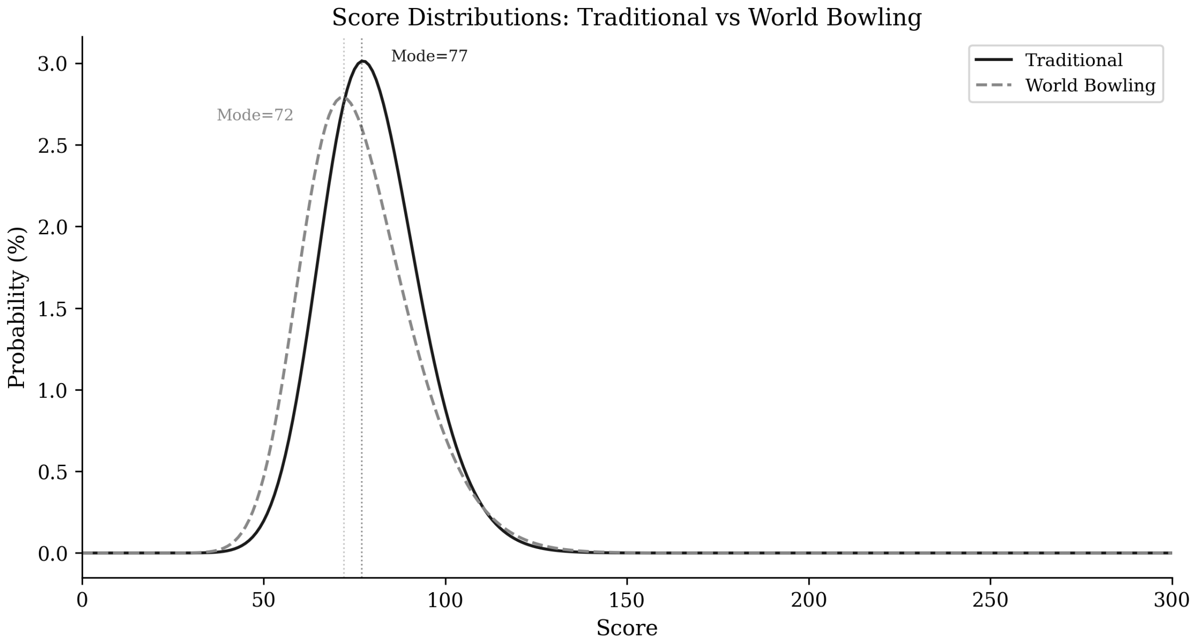

Figure 1: Score distributions under traditional and World Bowling scoring, computed by exact combinatorial enumeration over all possible games. Dashed lines indicate modes.

3.2 Impossible Scores

A structural property of World Bowling scoring is the impossibility of scores in the range 290–299. This arises because 9 strikes yield at most $9 \times 30 + 19 = 289$ (9 strikes plus one maximum spare of 10 + 9), while 10 strikes yield exactly 300. No sequence of legal balls can produce a game score between 290 and 299 inclusive under World Bowling rules. This gap is not a curiosity; it is a direct consequence of the linear, frame-independent scoring structure, and it reveals a fundamental property: **the score space is not fully utilised**.

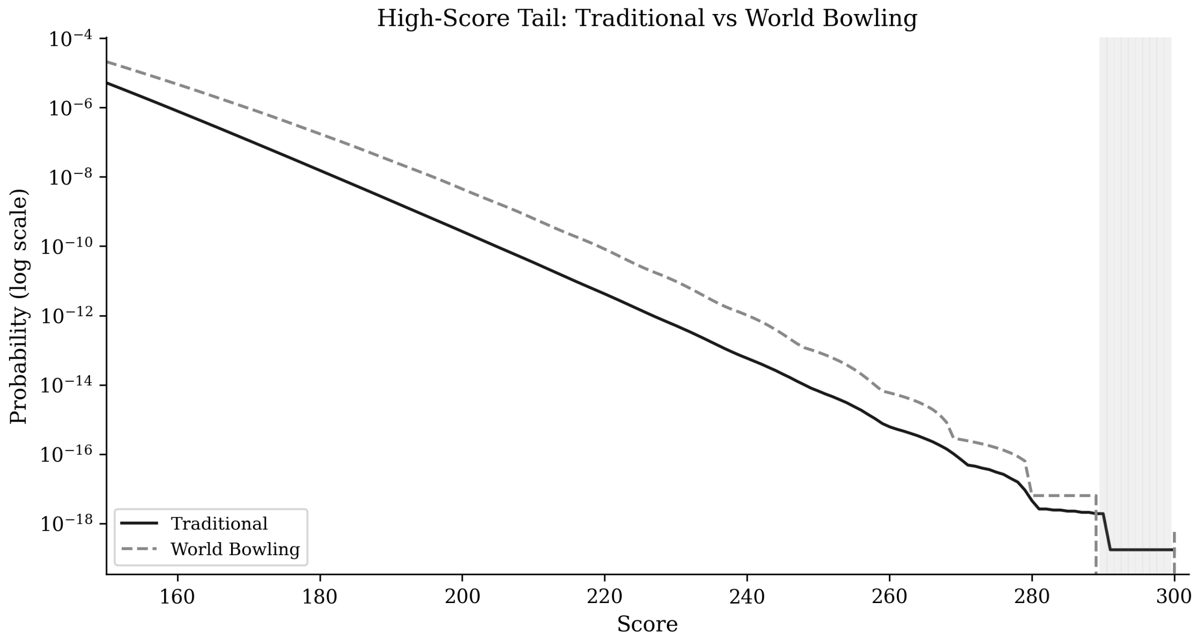

Figure 2: High-score tail on log scale. The shaded region at 290–299 marks scores that are mathematically impossible under World Bowling.

3.3 High-Score Compression

The number of distinct game sequences producing each high score differs markedly between systems:

Table 2: Number of distinct game sequences producing each high score.

Score	Traditional	World Bowling
200	1,526,313,637	7,019,752,752
240	335,065	1,599,447
260	3,534	9,225
280	26	10
290	11	0 (impossible)
300	1	1

At scores of 200 and above, World Bowling consistently offers more paths to the same score. This score compression means the score scale is coarser at the top: more distinct games collapse onto the same number, leaving fewer available values to separate high-scoring performances. Whether a coarser scale also reduces the system's ability to discriminate *player skill*, as opposed to merely separating individual games, is a separate question that the raw counts here cannot settle; we address it directly with a signal-to-noise reliability analysis in Part 2.

4 Sequence Sensitivity

4.1 Commutativity of World Bowling Scoring

Proposition 1. *Under World Bowling scoring, the total game score is invariant under any permutation of the ten frames.*

Proof. Let f_i denote the score of frame i under World Bowling rules. The total game score is $S = \sum_{i=1}^{10} f_i$. Since each f_i is computed from balls thrown within frame i alone, and addition is commutative and associative, any permutation σ of the frames yields $S' = \sum_{i=1}^{10} f_{\sigma(i)} = S$. □

The proposition is deliberately elementary: the proof is commutativity of addition, and no deeper machinery is needed. We state it formally because it is the structural anchor for everything that follows. The contribution of this paper is not the proof itself but the quantification, in Part 2, of exactly what a scoring system gives up by having this property.

This property, commutativity with respect to frame order, does not hold for traditional scoring. Under traditional scoring, a strike in frame i depends on balls thrown in frames $i+1$ and potentially $i+2$. Reordering frames changes which balls serve as bonuses, and therefore changes the score.

4.2 Empirical Demonstration

We compute scores for fixed frame compositions under varying orderings. Consider 9 frames comprising 5 strikes and 4 spares (5,5) in various orderings, followed by a neutral frame 10 (5, 4):

Table 3: Scores for four orderings of the same frame composition (5 strikes + 4 spares).

Sequence (9 frames + neutral 10th)	Traditional	World Bowling
5 strikes then 4 spares	204	219
Alternating: strike, spare × 4, strike	188	219
4 spares then 5 strikes	208	219
X, sp, sp, X, sp, X, X, sp, X	188	219

Under World Bowling, all four orderings score identically (219). Under traditional scoring, the score varies by 20 points (188–208) depending solely on the ordering.

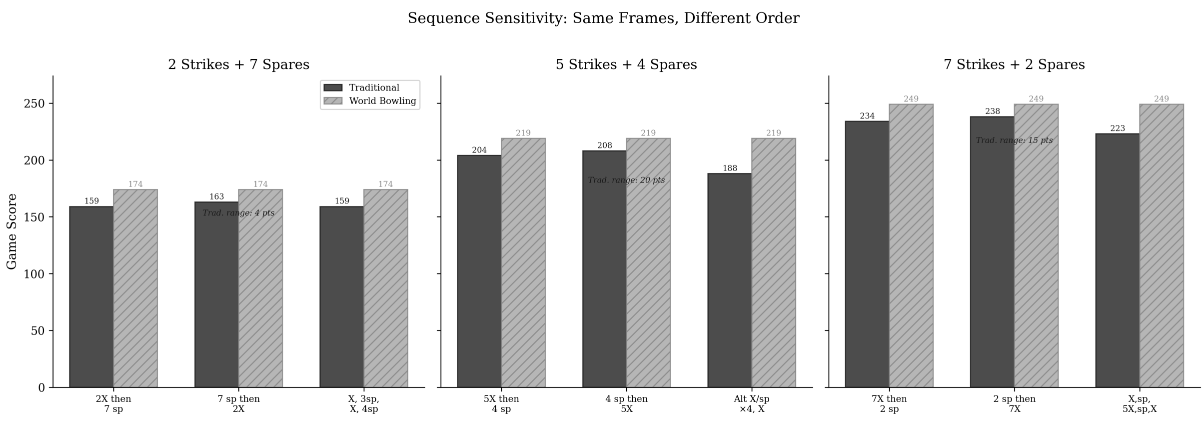

Figure 3: Sequence sensitivity across three compositions (2, 5, and 7 strikes out of 9 frames). In each panel, the same frames are reordered. World Bowling scores are identical for all orderings; traditional scores vary, with the range increasing at higher strike counts.

4.3 Exhaustive Permutation Analysis

To quantify the full extent of sequence sensitivity, we enumerate all distinct permutations of fixed frame compositions and compute the score under each system. For compositions of k strikes and $(9-k)$ spares in frames 1–9:

Table 4: Exhaustive permutation analysis. Traditional score range and standard deviation grow with the mix of frame types; World Bowling range is exactly zero for every composition.

Composition	Orderings	Trad. range	Trad. SD	WB range
1 strike + 8 spares	9	5	1.7	0
2 strikes + 7 spares	36	6	2.1	0
3 strikes + 6 spares	84	11	2.9	0
4 strikes + 5 spares	126	16	4.2	0
5 strikes + 4 spares	126	21	5.5	0
6 strikes + 3 spares	84	26	6.4	0
7 strikes + 2 spares	36	26	6.6	0
8 strikes + 1 spare	9	15	5.6	0

The World Bowling score range is exactly zero for every composition, confirming Proposition 1 empirically. The traditional score range grows with the mix of frame types, peaking at 26 points for compositions with 6–7 strikes. When open frames are included alongside strikes, the range increases further: 5 strikes mixed with 4 open frames (5, 4) produces a 39-point range across 126 orderings.

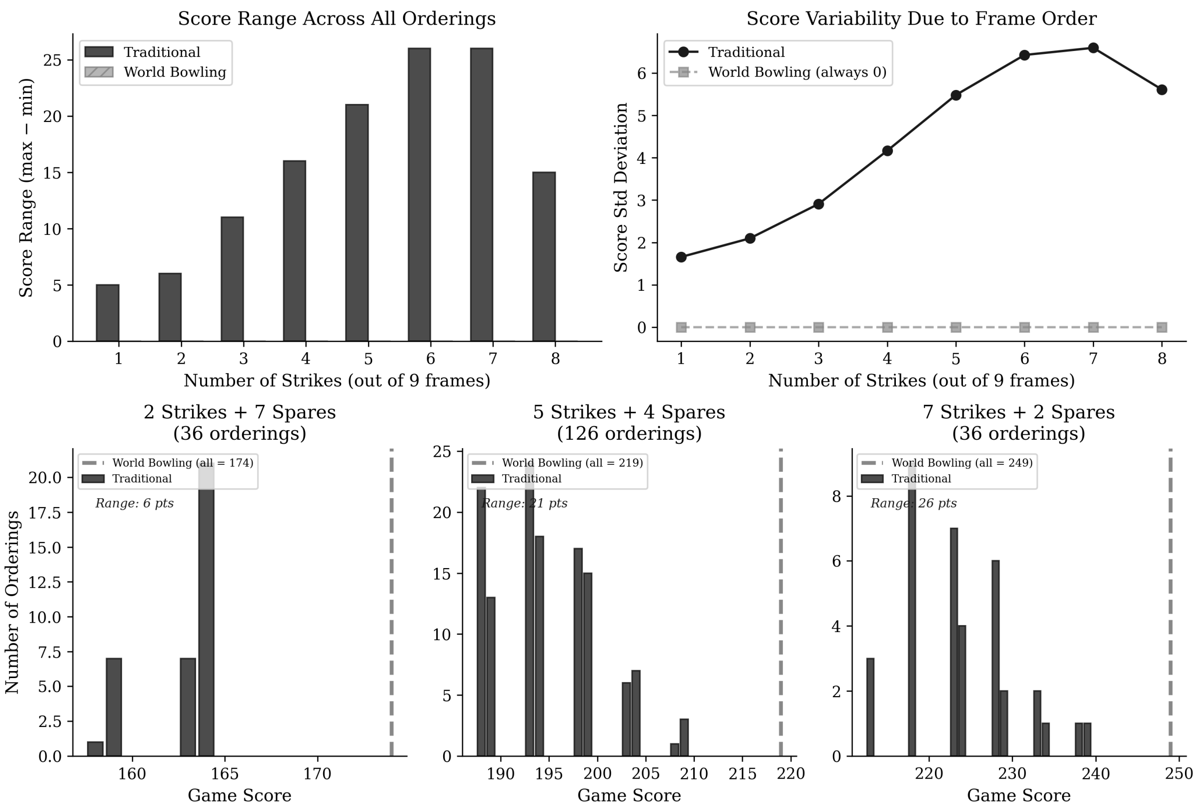

Figure 4: Sequence sensitivity of fixed frame compositions. Top: score range and score standard deviation across all orderings, by composition. Bottom: score distribution across all orderings for three compositions (2, 5, and 7 strikes). Traditional scores spread across a range that widens with strike count; World Bowling is exactly zero throughout and collapses to a single value (dashed line).

4.4 Implications for Skill

Sequence sensitivity has a direct sporting interpretation. In competitive bowling, a player who strings consecutive strikes together is demonstrating a qualitatively different level of performance than one who scatters the same number of strikes throughout a game. The traditional scoring system encodes this distinction numerically. The World Bowling system does not.

The maximum score achievable with a fixed number of strikes is higher when those strikes are consecutive (due to compounding bonuses from subsequent strikes), creating a score incentive for sustained excellence. Under World Bowling, a bowler who throws five consecutive strikes in frames 1–5 and then opens all remaining frames achieves the same score as one who alternates strikes and open frames, regardless of how much more difficult the sustained streak was to achieve.

5 The Reward Gradient for Consecutive Strikes

We define the **marginal strike value** as the increase in game score when one additional open frame (5, 4) is replaced by a strike, all other frames remaining constant. Under World Bowling this value is trivially constant: $30 - 9 = 21$ points per strike added.

Under traditional scoring, the marginal value depends on the surrounding context:

Table 5: Marginal value of each additional consecutive strike.

Consecutive strikes	Trad. Score	World Score	Trad. Marginal	World Marginal
0	90	90	n/a	n/a
1	100	111	+10	+21
2	116	132	+16	+21
3	137	153	+21	+21
4	158	174	+21	+21
5	179	195	+21	+21
10	300	300	+37	+21

Two observations are immediate. First, the initial strikes are undervalued in traditional scoring relative to World Bowling: the first strike adds only 10 points compared to 21 under World Bowling. This is because the bonus from a lone strike depends on subsequent balls that are in this model open frames (5, 4 = 9 pins), capping the bonus at 9. Second, the final strike in a perfect game is worth 37 points under traditional scoring, substantially more than World Bowling's flat 21, because it is preceded by two strikes that each earn its bonus retroactively.

This structure rewards the *completion* of a skill streak more than its initiation. The first strike of a sequence is worth relatively little; the strike that extends a string is worth substantially more. Under World Bowling, every strike is worth exactly the same regardless of what surrounds it.

Part 2: Skill-Weighted Simulation

6 Simulation Model

6.1 Player Model

To ground the mathematical analysis in realistic playing conditions, we simulate bowling games across six empirically motivated skill tiers. Each player is characterised by three parameters:

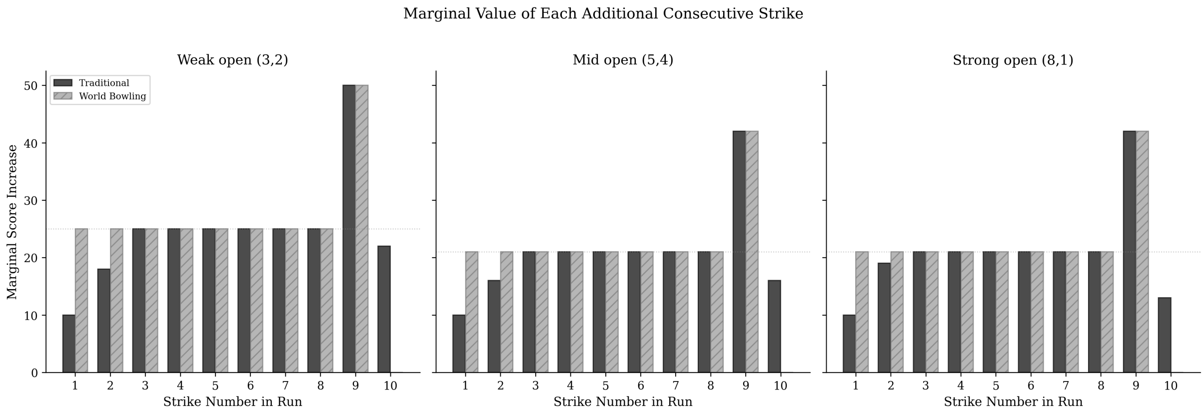

Figure 5: Marginal value of each additional consecutive strike, shown for three different non-strike fill patterns: weak (3,2), mid (5,4), and strong (8,1). The compounding pattern in traditional scoring and the flat World Bowling marginal are structural, not dependent on the fill choice.

strike probability (p_s), spare conversion probability (p_{sp}), and mean first-ball pin count when not striking. Tier parameters were calibrated against published data from multiple sources: McCarthy (2011) reports a mean score of 205.65 (SD 31.91) across 278,579 PBA National Tour games, with individual bowler ability estimates ranging from approximately 150 to 235; VanDerwerken & Kenter (2018) reports posterior mean strike probabilities ranging from approximately 53% to 66% across 53 PBA bowlers; and USBC published league averages anchor the recreational end.

Table 6: Skill tier parameters. The recreational tier targets casual bowlers based on USBC league averages. The professional tier targets the PBA Tour season average of approximately 230, consistent with the upper range of bowler ability estimates reported by McCarthy (2011). The Top 10 tier (73% strike rate) is extrapolated from the upper tail of the professional ability distribution, representing the elite end of PBA performance (VanDerwerken & Kenter, 2018; Yaari & Eisenmann, 2012).

Skill Tier	Strike Rate	Spare Rate	Pin Mean	Target Trad. Mean
Recreational	4%	27%	4.2	~90
Club	20%	47%	5.5	~130
Competitive	40%	64%	6.5	~175
Elite	55%	77%	7.2	~210
Professional	66%	86%	7.8	~230
Top 10	73%	92%	8.2	~245

6.2 Simulation Protocol

We employ three progressively realistic simulation models to ensure robustness of results:

1. **Independent model (base case).** Each ball is generated independently: the first ball of each frame is a strike with probability p_s, otherwise pins follow a binomial distribution; the second ball converts the spare with flat probability p_{sp}.
2. **Momentum model.** Strike probability increases by 5 percentage points after a strike and decreases after an open frame, capturing the simplest form of sequential dependence.
3. **Markov chain model with spare difficulty.** Strike probability follows a first-order Markov chain with state-dependent transition probabilities informed by the empirical hot-hand magnitudes reported in VanDerwerken & Kenter (2018), who built on the streak

probability framework of Martin (2006) to fit 4th-order Markov models to 26,102 PBA tournament games from 53 professional bowlers. Their posterior estimates show strike probability increasing by approximately 5–10 percentage points after consecutive strikes relative to a cold state. We apply conservative first-order transition ratios: $P(\text{strike} \mid \text{prev. strike}) = p_s \times 1.15$, $P(\text{strike} \mid \text{prev. spare}) = p_s \times 0.95$, $P(\text{strike} \mid \text{prev. open}) = p_s \times 0.85$. Spare conversion is leave-dependent, with three difficulty classes consistent with the non-strike leave distributions reported in VanDerwerken & Kenter (2018) Tables 7–8: easy single-pin leaves (55% of non-strike results, high conversion), moderate multi-pin leaves (35%, baseline conversion), and splits (10%, conversion rate reduced to 30% of baseline).

For each skill tier, we simulate 50,000 complete games under each model. Every simulated game is scored under both traditional and World Bowling rules, enabling paired comparison. The independent model provides a conservative lower bound; the Markov model provides the most realistic estimate.

6.3 Tier Results

Table 7: Simulated score statistics by skill tier (50,000 games each).

Skill Tier	Traditional Mean (SD)	World Bowling Mean (SD)
Recreational	91 (14.2)	95 (17.7)
Club	133 (21.2)	149 (26.8)
Competitive	176 (25.8)	200 (28.1)
Elite	209 (26.6)	233 (25.1)
Professional	232 (25.7)	253 (21.4)
Top 10	246 (24.4)	265 (18.6)

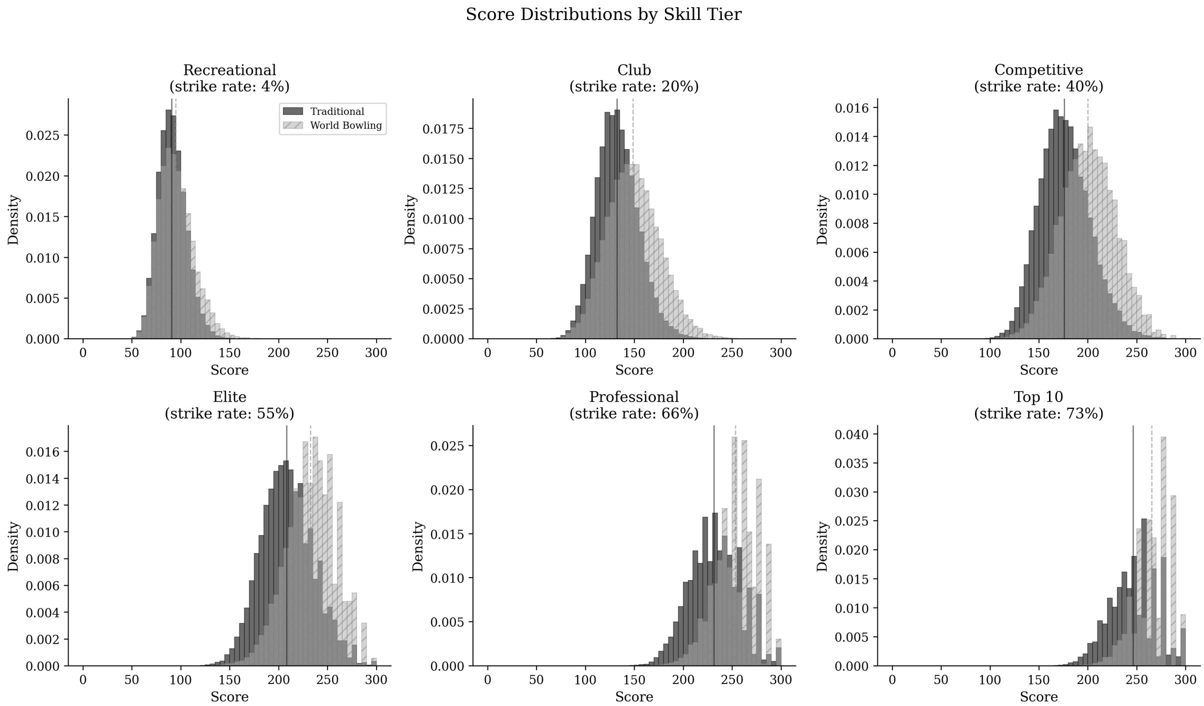

Figure 6: Score distributions at each skill tier under both scoring systems.

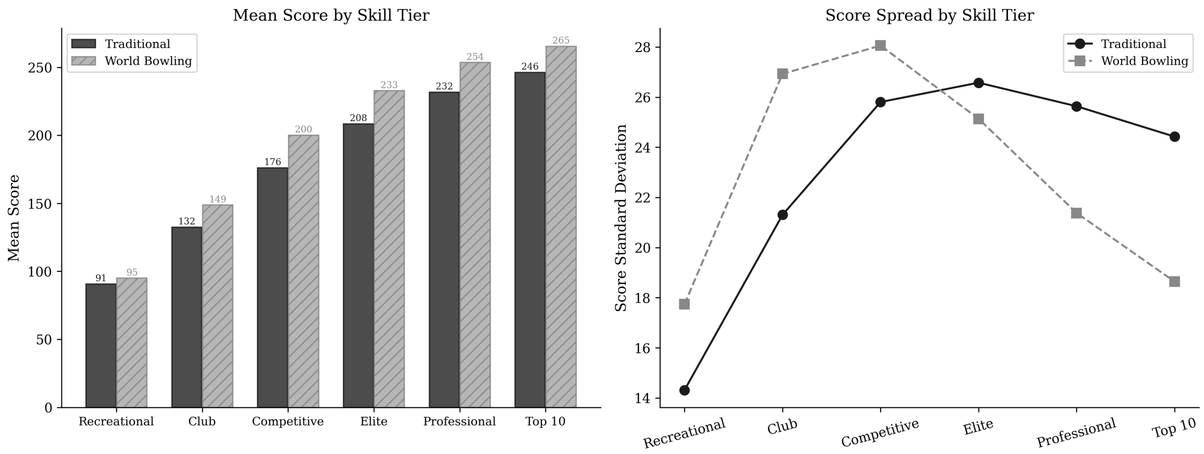

Figure 7: Left: mean scores by skill tier. Right: score standard deviation (note the crossover between systems).

7 The Score Spread Crossover

7.1 Standard Deviation as a Function of Skill

A critical observation emerges from the simulation: the standard deviation of scores under each system behaves differently as a function of player skill.

Under World Bowling, score standard deviation peaks at low-to-moderate skill levels (SD $\approx$ 28 at competitive level) and then **decreases monotonically** as skill increases, reaching SD $\approx$ 18.6 at the Top 10 level. This compression occurs because high-skill World Bowling games converge toward the maximum of 300. With most frames scoring 30 (strike), the remaining variance comes only from occasional non-strikes.

Under traditional scoring, standard deviation increases with skill through the competitive range and remains elevated (SD $\approx$ 25.7 at professional, $\approx$ 24.4 at Top 10). This is because the compounding bonus structure amplifies the consequences of each non-strike: a single open frame in an otherwise perfect game costs substantially more under traditional scoring than under World Bowling, maintaining score spread even at elite levels.

7.2 The Crossover Point

We sweep strike rate continuously from 10% to 80% and compute score standard deviation under each system. The two curves cross at a strike rate of **48.7%** (95% bootstrap confidence interval 48.2%–49.1%, from 500 resamples of the simulated games). Below this threshold, World Bowling produces greater score spread; above it, traditional scoring produces greater spread.

It is tempting to read this crossover as a crossover in *discrimination*, with World Bowling separating players better below 50% and traditional scoring better above it. That reading is wrong, and correcting it is the central methodological point of this section. Standard deviation measures how wide the score distribution is, not how much of that width reflects genuine differences in player ability. A scoring system can inflate its standard deviation purely by amplifying the consequences of a single lucky or unlucky ball, which is noise, not skill. The next two subsections decompose the spread into signal and noise and show that, on the axis of raw skill, the extra spread traditional scoring carries at elite levels is mostly the latter. The crossover is best understood as a statement about score *granularity and margins*, the size of the gaps between scores, rather than about how reliably a single game ranks two players.

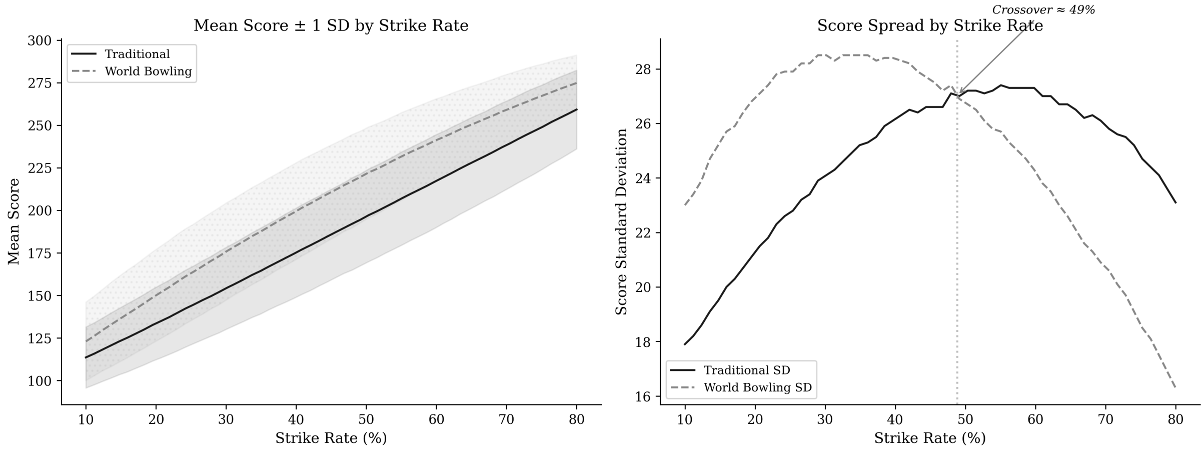

Figure 8: Left: mean score $\pm$ 1 SD by strike rate under both systems. Right: score standard deviation as a function of strike rate, showing the crossover at approximately 49% (bootstrap 95% CI 48.2%–49.1%). Note that this is a crossover in score spread, not in discrimination of skill; see Figure 9.

7.3 Signal versus Noise: A Reliability Analysis

The standard deviation of game scores conflates two very different things: genuine, persistent differences between players (signal) and game-to-game variation for a single fixed player (noise). A scoring system that simply magnifies the score impact of a single ball will show a large standard deviation without discriminating skill any better. To separate the two, we treat each scoring system as a measurement instrument and estimate its single-game reliability.

For a target skill level, we construct a population of players whose true strike rates differ by a realistic amount (a standard deviation of three strike-rate points, reflecting the within-neighbourhood ability spread among players nominally at the same level; cf. the roughly 53%–66% posterior strike-rate range across 53 PBA bowlers in VanDerwerken & Kenter (2018)). We simulate many games per player, score every game under both systems from the *same* ball sequences, and decompose the total score variance into a between-player component σ_B^2 and a within-player component σ_W^2. The single-game reliability is the intraclass correlation $\mathrm{ICC} = \sigma_B^2/(\sigma_B^2 + \sigma_W^2)$, the single-measurement ICC under a one-way random-effects model: the fraction of one game's score variance attributable to real skill differences. A higher ICC means a single game more reliably identifies the better player. The absolute level of the ICC depends on the heterogeneity of the reference population (here a three-point standard deviation in true strike rate), so it is the comparison between systems, not the level, that is the result of interest; the small remaining point-to-point variation in Figure 9 is Monte Carlo error.

Two findings emerge (Figure 9). First, the two systems have nearly identical single-game reliability at every skill level: the ICC estimates differ by less than 0.01 at strike rates above roughly 15%, and by at most 0.025 at the lowest recreational levels, where World Bowling is if anything marginally the more reliable measure. On the axis of raw skill, traditional scoring is no better at separating players than World Bowling. Second, the variance decomposition explains why traditional scoring's larger elite-level spread does not buy discrimination: at the top of the range, almost all of the extra spread is noise. At a 78% strike rate, traditional scoring's within-player (noise) standard deviation is about 24 points against World Bowling's 17, while the between-player (signal) standard deviations are comparable and small (roughly 6 versus 4 points). The amplifying bonus structure inflates noise and signal together, leaving their ratio, and hence reliability, essentially unchanged.

This result aligns with the between-tier comparison: across adjacent skill tiers, both systems produce nearly identical Cohen's d effect sizes, from 2.3 (recreational vs club) down to 0.6 (professional vs Top 10), matching to within 0.03 at every boundary. Whether one measures discrimination between tiers or reliability within a population, the two systems are equivalent on raw skill. The case for traditional scoring cannot rest on the standard-deviation crossover, and we do not rest it there.

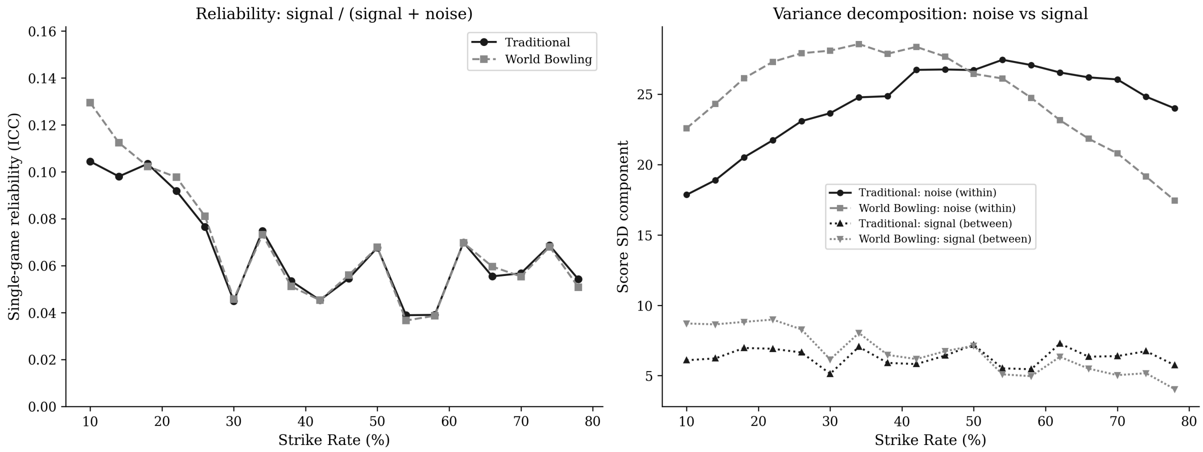

Figure 9: Left: single-game reliability (ICC) as a function of strike rate. The two systems track each other across the entire skill range. Right: the variance decomposition. At elite levels the within-player noise component diverges (traditional scoring is noisier) while the between-player signal component stays comparable, so traditional scoring's larger spread is mostly amplified noise.

7.4 Discriminating Streakiness at Fixed Skill

If the two systems are equally reliable on raw skill, where does commutativity actually cost something? Precisely where Proposition 1 predicts: in measuring *how* a player arranges a fixed number of strikes. To isolate this, we hold the marginal strike rate constant and vary only the lag-1 autocorrelation ϕ of the strike indicator, the tendency to throw strikes in runs rather than scattering them. A two-state Markov chain lets us set any autocorrelation while keeping the expected strike count fixed, so two players with different ϕ are identical in raw skill and differ only in streakiness. The magnitude range we sweep (ϕ up to 0.35) spans the hot-hand effect sizes reported for professional bowlers by VanDerwerken & Kenter (2018).

The contrast is stark (Figure 10). Across a population of players who share a 66% strike rate but differ in streakiness, a player's mean traditional score correlates with true streakiness at $r = 0.49$, while the World Bowling correlation is $r = -0.04$, indistinguishable from zero. Comparing the extremes directly, a maximally streaky player ($\phi = 0.35$) outscores a streak-free player ($\phi = 0$) of identical raw skill by about seven points under traditional scoring (235.5 vs 228.6; Cohen's $d = 0.23$), but by nothing under World Bowling (250.6 vs 251.2; $d = -0.02$). World Bowling is not merely noisier at detecting streakiness; it is structurally incapable of detecting it, exactly as commutativity requires. This, and not the spread crossover, is the discrimination that current-frame scoring discards.

The effect is specific to strikes. Repeating the experiment on the spare dimension, holding the strike process independent and instead clustering spare *conversions* at a fixed marginal conversion rate, produces no signal under either system (traditional $r = 0.02$, World Bowling $r = 0.01$). Part of this null is mechanical: at elite conversion rates of 86–92% there is little variance left to cluster. But the structural reason is decisive regardless: a strike's bonus looks two balls ahead

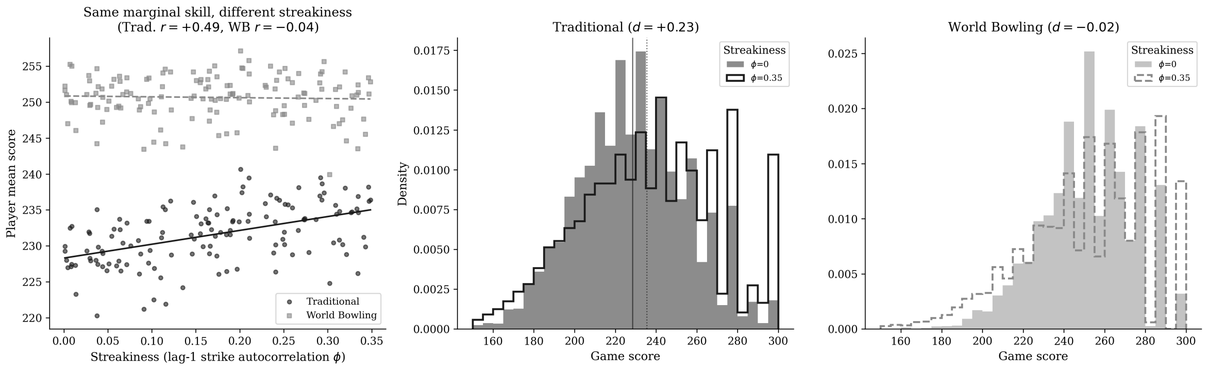

Figure 10: Left: player mean score against true streakiness (ϕ) for a population of players who share an identical 66% strike rate. Traditional scores rise with streakiness; World Bowling scores are flat. Middle and right: game-score distributions for a streak-free and a maximally streaky player of identical raw skill, under traditional scoring and World Bowling respectively. Traditional scoring shifts; World Bowling does not.

and can chain into a following strike, so consecutive strikes compound, whereas a spare's bonus looks only one ball ahead and never chains into another bonus. Spare conversion is a genuine skill, and both systems measure it about equally; but only the *arrangement* of strikes, not of spares, carries sequencing information, and only traditional scoring records it.

7.5 Compounding over a Series

The per-game streakiness effect is real but moderate ($d = 0.23$). Competition, however, is decided over multi-game blocks and tournaments, and an independent per-game advantage compounds: the expected total-score gap grows linearly in the number of games while its standard deviation grows only as the square root, so the series-level effect size grows as $\sqrt{n}$. Figure 11 traces this for our two identical-skill players, one streak-free and one streaky. Under traditional scoring the streaky player's probability of winning a head-to-head series rises from 57% over a single game to 67% over a six-game block and 90% over a 48-game tournament, and the series effect size grows from $d = 0.26$ to $d = 1.79$. Under World Bowling the same player's win probability never leaves the neighbourhood of chance (53% even at 48 games). These figures should be read as upper bounds on the practical stakes: they contrast the extremes of the swept streakiness range ($\phi = 0.35$ versus $\phi = 0$), and the between-player dispersion of streakiness in real professional populations is not yet established. If the population spread of ϕ is narrower than the range swept here, the win probabilities move toward chance proportionally. Subject to that caveat, a property that looks minor in a single game becomes, over the span of a real competition, frequently decisive under traditional scoring and nearly invisible under World Bowling.

8 Professional-Level Sequence Analysis

8.1 Specific Sequence Comparisons

To illustrate the practical consequences at the professional level, we compute scores for sequences representative of elite play:

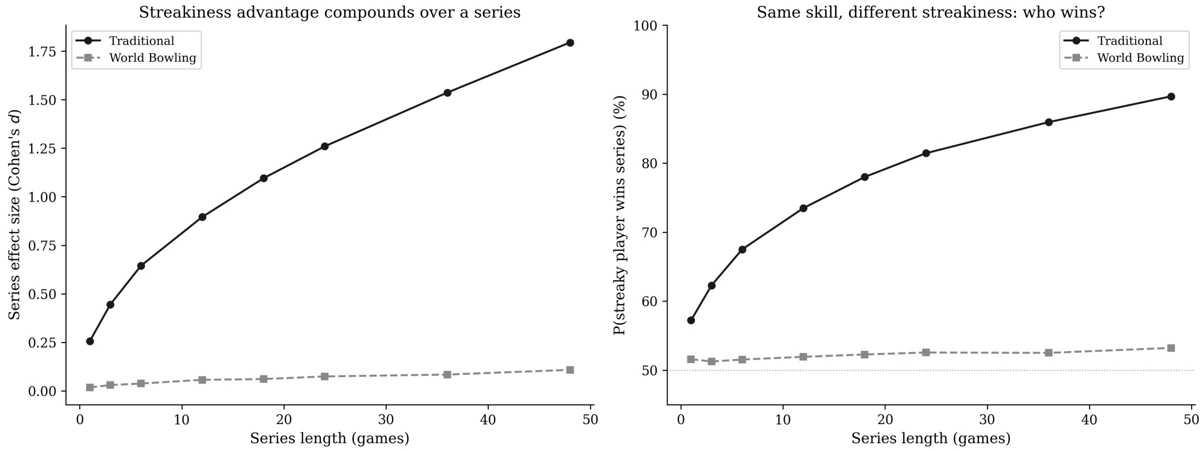

Figure 11: Left: the series-level effect size (Cohen's d) separating a streaky player from a streak-free player of identical 66% strike rate, as a function of series length; it grows as $\sqrt{n}$ under traditional scoring and stays near zero under World Bowling. Right: the probability that the streaky player wins a head-to-head series of the given length.

Table 8: Professional-level sequence comparisons. The systems agree at the extremes but diverge substantially for mixed sequences.

Sequence	Traditional	World Bowling	Difference
Perfect game (12 strikes)	300	300	0
10 strikes + spare(7,3) + 7	287	300	−13
8 strikes + 2 spares(7,3) + 7	261	274	−13
6 spares then 4 strikes + X,X	215	210	+5
4 strikes then 6 spares + 5	195	210	−15
Alternating strike/spare × 5	200	225	−25
All spares (5,5) + 5	150	150	0

Two patterns are notable. First, the systems agree at the extremes (perfect game and all spares) but diverge substantially for the mixed sequences that constitute the vast majority of professional games. Second, the *direction* of the difference depends on the sequence: traditional scoring rewards sequences with consecutive strikes more than World Bowling does, while World Bowling rewards isolated strikes more. The traditional system encodes *how* the strikes were arranged; World Bowling treats them as interchangeable.

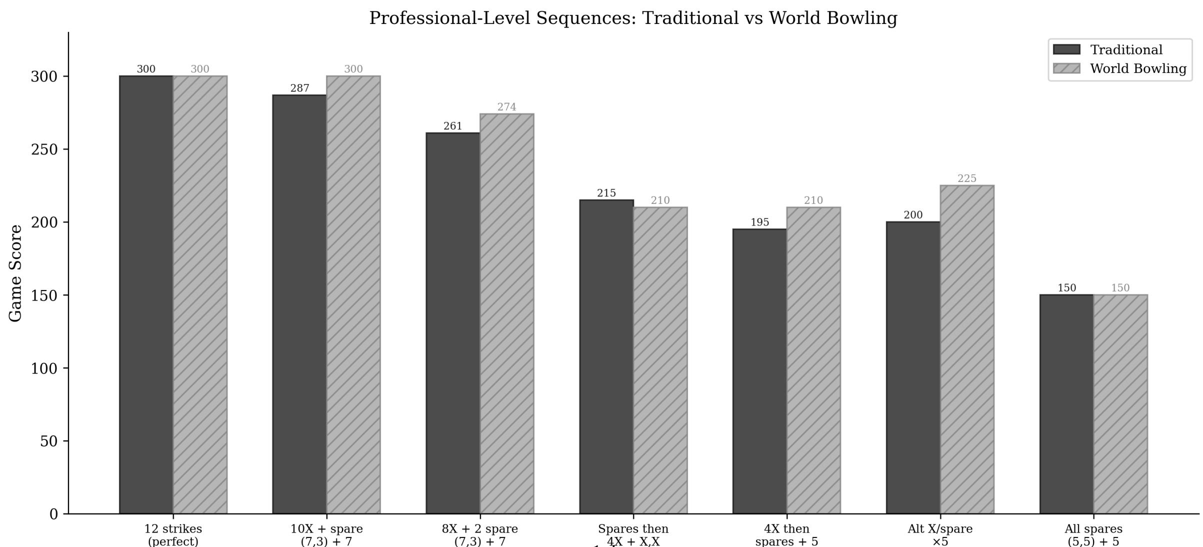

Figure 12: Professional-level sequence comparison under both scoring systems.

9 Discussion

9.1 What Scoring Systems Measure

A scoring system is a function from a sequence of physical performances to a number. The properties of that function determine what the number communicates about the athlete's skill. This perspective connects to the broader theory of proper scoring rules, where the design of a measurement function determines whether it faithfully elicits the quantity it claims to measure (Gneiting & Raftery, 2007). Two scoring systems can agree on the maximum possible score (both systems peak at 300 for a perfect game) while disagreeing significantly on how they encode the path to that maximum.

Traditional bowling scoring encodes two distinct performance qualities simultaneously: the absolute pin count (how many pins were knocked down) and the sequencing of that performance (whether those knockdowns were concentrated in consecutive frames or distributed across the game). World Bowling scoring encodes only the first.

9.2 The Commutativity Trade-off

The commutativity of World Bowling scoring (Proposition 1) is its defining feature. It makes the system simple to compute and explain. But commutativity in a scoring function means that sustained performance is not valued above distributed performance. In most competitive sports, consecutive success is considered more difficult than the same success spread over time. A tennis player who wins six consecutive games demonstrates something different from one who wins six non-consecutive games. Traditional bowling scoring agrees with this intuition. World Bowling scoring does not.

9.3 Is Streakiness a Skill, and Should It Be Rewarded?

Our central empirical claim is that traditional scoring rewards streakiness and World Bowling does not. Whether that is a virtue depends on a question our data cannot fully settle: is the tendency to throw consecutive strikes a stable, skill-based attribute, or an artefact of luck and changing conditions? The hot-hand question, whether streaks reflect a causal change in player state or merely the perception of one (Gilovich et al., 1985), is pointed on exactly this in bowling. Dorsey-Palmateer & Smith (2004) find positive autocorrelation in strike sequences, but Yaari & Eisenmann (2012), analysing a large body of data, conclude that the correlation reflects association with recent results rather than a causal "hot hand," and is plausibly driven by varying lane conditions, equipment, and form across a session rather than by momentum within a player. We take this seriously, because it bears directly on whether rewarding streaks rewards skill or amplifies noise.

We make three points in response. First, our finding is robust to the interpretation: regardless of *why* strikes cluster, traditional scoring records the clustering and World Bowling discards it. The structural fact stands. Second, even on the conditions-based reading, the within-game arrangement of strikes is not purely random with respect to the player. A bowler who reads a lane, adjusts, and converts a favourable stretch into a run of strikes is exercising a competitive capability, even if "momentum" is the wrong mechanism for it; the relevant question for a scoring system is whether the achievement happened, not which latent process produced it. Third, and most importantly, whether sequencing *ought* to count is a normative choice about what the sport wishes to measure, not a fact to be discovered. Our contribution is to make that choice explicit and quantified: traditional scoring treats sequencing as part of performance, World Bowling treats it as irrelevant, and a governing body should choose between them knowing precisely which property it is keeping or discarding. We note the streakiness signal we measure ($r = 0.49$

at the population level, $d = 0.23$ between extremes) is real but moderate, so the practical stakes of this choice, while genuine, should not be overstated.

9.4 The Appropriate Domain for Each System

We do not argue that current-frame scoring lacks merit. For recreational participants, its simplicity and accessibility represent genuine improvements over traditional scoring. The simulation results support this: below approximately 50% strike rate, World Bowling produces *greater* score spread, and its simpler computation removes a barrier to casual participation. The higher Shannon entropy of the World Bowling distribution (5.94 vs 5.81 bits; Shannon (1948)) reflects this wider spread, since entropy here measures dispersion rather than discrimination, a distinction our reliability analysis makes concrete. A casual player's score therefore more directly reflects their raw pin count without the amplifying effect of bonus compounding.

Our argument is narrower, and the reliability analysis sharpens it. At the competitive and professional levels, where strike rates exceed 50% and the ability to string consecutive strikes is a distinguishing skill, current-frame scoring is not a worse measure of raw skill, but it is blind to the sequencing dimension that traditional scoring records (Figure 10). The spread crossover at approximately 49% strike rate is not the basis of this argument; it marks where the score scale changes character, not where discrimination of skill reverses.

Nor is the choice binary. Traditional and current-frame scoring are two points in a design space, not the whole space. A governing body that valued both accessibility and sequencing could adopt a hybrid: current-frame base scoring with an explicit consecutive-strike bonus (for example, a fixed supplement for a double or turkey) would recover much of the sequencing signal at a fraction of traditional scoring's computational complexity. Our framework supplies the tools to evaluate such designs: enumerate the distribution, measure the sequence sensitivity, and estimate the reliability before changing the rules. Mapping that design space is a natural extension of this work.

9.5 Score Compression at the Elite Level

The high-score compression shown in Table 2, combined with the decreasing standard deviation under World Bowling at high skill levels (Figure 8), narrows the score range available to separate elite players. At the Top 10 professional level, World Bowling's score SD of 18.6 compared to traditional's 24.4 means roughly 24% less score spread, with a similar gap at the professional tier (SD 21.4 vs 25.7). It is important to state precisely what this does and does not imply. As the reliability analysis showed (Figure 9), the wider traditional spread is mostly amplified noise, so the two systems rank players about equally well in any single game; the compression does not, by itself, make World Bowling a worse discriminator of raw skill. What compression does affect is the granularity of the score scale: with scores packed into a narrower band and 290–299 unreachable, more distinct performances collapse onto the same number, leaving fewer score values to register fine differences and break ties over a tournament. The substantive loss is not resolution of raw skill but resolution of *sequencing*, which Figure 10 shows traditional scoring records and World Bowling cannot.

9.6 End-Loading and the Perception of Unfairness

Sequence sensitivity has a second, less obvious asymmetry beyond the reward gradient: under traditional scoring, a ball thrown late in the game contributes more to the total than the same ball thrown early. Every ball enters the score up to three times, once in its own frame and once in each of up to two preceding frames as a bonus, so the balls of frame 1 enter exactly one frame-score while the balls of frame 9 can enter three. A weak frame is correspondingly cheaper early than late. Holding the frame composition fixed, a player who throws nine strikes and one spare (5,5) scores 290 if the spare comes first but 270 if it comes last; with nine strikes and one open frame (9,0), the scores are 279 and 267. Under World Bowling both orderings score identically (285 and 279 respectively). The tenth-frame open is, structurally, the most expensive mistake in traditional scoring.

This end-loading is the source of a recurring informal complaint about traditional scoring: two players who throw identical frames in different orders receive different scores, and the difference is not symmetric, because the system forgives a slow start more readily than a weak finish. What the complaint identifies is not a flaw but a design property. Traditional scoring places a premium on the closing frames, much as other sports reward performance in high-leverage moments. Whether this clutch premium is desirable, whether the sport wants the tenth frame to matter most, is a normative question, and it is the same question the sequencing result poses from another angle: current-frame scoring is neutral to order, treating a strike in frame 1 and a strike in frame 10 as identical events.

The hot-hand literature, extensively reviewed by Bar-Eli et al. (2006) and investigated in bowling specifically by Yaari & Eisenmann (2012) and Dorsey-Palmateer & Smith (2004), documents that strike sequences are autocorrelated rather than independent, though it disputes whether the cause is a causal hot hand or shared dependence on session conditions. Either way, both the sequencing and the end-loading exist in the scoring record; whether a scoring system should reflect them is the normative question discussed above.

9.7 The 290–299 Gap as a Design Artefact

The impossibility of World Bowling scores in the range 290–299 is mathematically unavoidable given the system's design. It is a direct consequence of the discrete jump between the maximum achievable score with 9 strikes (289) and the minimum score with 10 strikes (300). This gap means that a bowler who throws nine strikes and one near-perfect spare cannot be scored above 289, while one who converts that spare to a tenth strike jumps directly to 300. The scoring cliff at the top of the range is not a feature of athletic performance; it is an artefact of linear frame independence.

9.8 Robustness Across Simulation Models

Our base simulation assumes frame-to-frame independence: each ball's outcome depends only on the player's fixed skill parameters, not on preceding results. This is deliberately conservative. The hot hand literature in bowling (Dorsey-Palmateer & Smith, 2004; Yaari & Eisenmann, 2012) provides empirical evidence that consecutive strikes are positively correlated (a "momentum" or streakiness effect) which an independence model ignores.

To test the sensitivity of our conclusions to this assumption, we compare three simulation models of increasing realism: the independent base case, a simple momentum model ($\pm 5\%$ strike probability shift after strikes/opens), and a first-order Markov chain with leave-dependent spare difficulty (Section 6.2). Figure 13 shows the score standard deviations and the SD gap (traditional minus World Bowling) across all six skill tiers under each model. Two observations are immediate. First, all three models agree on the *direction* of the effect at every tier: World Bowling provides more spread below competitive level, and traditional scoring provides more spread above it. The crossover point is robust to model choice. Second, the more realistic models produce *larger* gaps at the professional and Top 10 levels: the Markov model yields a traditional scoring advantage of +5.8 SD points at the Top 10 tier, compared to +4.1 under independence. The independence assumption is therefore conservative: any realistic model of sequential dependence amplifies the effect.

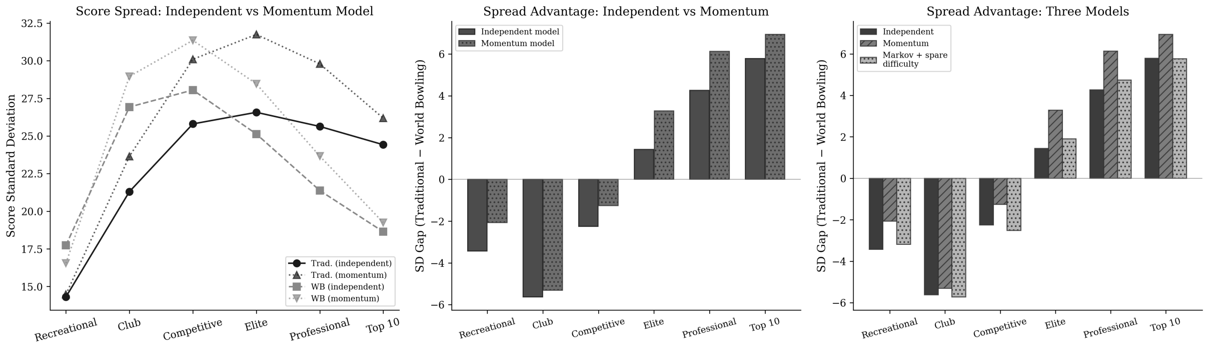

Figure 13: Robustness across simulation models. Left: score standard deviation by skill tier under independent and momentum models. Middle: SD gap (traditional minus World Bowling) by tier under each. Right: SD gap under all three models, including the Markov chain with spare difficulty. More realistic models amplify the traditional scoring advantage at elite tiers.

9.9 Limitations and Future Work

This paper has several limitations that suggest directions for future work. First, the simulation uses a parametric skill model rather than real game data. As a face-validity check, a population mixture of bowlers with a mean strike rate of 54% and an ability standard deviation of eight strike-rate points reproduces the pooled PBA moments reported by McCarthy (2011) closely: simulated mean 204.4 and standard deviation 31.5, against the published 205.65 and 31.91. This confirms that the model reproduces the real aggregate score distribution and not merely its mean, but it is not a substitute for ball-by-ball validation. We want to be precise about the scope of this limitation: the calibration validates the model's *marginal* behaviour (score means and spreads by tier), while the paper's headline claims concern *sequential* structure, and the sequential inputs, the Markov transition ratios and the spare leave-class probabilities, are assumed from published ranges rather than fitted to ball-by-ball data. Applying both scoring systems to a database of actual PBA tournament games, and re-estimating the reliability and streakiness results on real strike sequences, would provide a far stronger empirical anchor, and it is the most important extension of this work. Second, while the Markov chain model captures first-order sequential dependence, more sophisticated models incorporating fatigue, lane oil breakdown, and frame-specific strategy could further refine the magnitude estimates. Third, we show that the per-game sequencing effect compounds over a multi-game series under an independence assumption (Figure 11), but a fuller tournament-format treatment, incorporating within-session correlation between games (shared lane conditions and form), bracket structures, and qualifying cuts, would sharpen the estimate of how often sequencing actually decides real competitions. The direction of this effect is genuinely uncertain: if streakiness is a persistent player attribute, within-session correlation lets the streaky player's advantage persist across games, making our estimate conservative; but if streaks are driven by lane conditions shared with the opponent on the same pair, the *differential* between players could instead shrink. Fourth, a formal Win Probability Added (WPA) analysis, computing how each scoring system affects the probability of winning a head-to-head match at each game state, would complement the present score-distribution analysis with a competition-dynamics perspective. The source code accompanying this paper is designed to support these extensions.

10 Conclusion

We have demonstrated, through exact combinatorial analysis and skill-weighted simulation, that traditional and World Bowling scoring systems differ in mathematically precise and practically significant ways:

1. **Traditional scoring is sequence-sensitive.** The order of frames matters, and it matters in a way that rewards consecutive success. World Bowling scoring is commutative; order is irrelevant. Exhaustive permutation analysis shows that the same frame composition can produce a score range of up to 39 points under traditional scoring, while World Bowling collapses all orderings to a single score.

2. **Traditional scoring has a compounding reward gradient for consecutive strikes.** Each strike in a string is worth more than an isolated strike, creating a non-linear incentive for sustained excellence. World Bowling scoring is linear: every strike is worth the same regardless of context.

3. **On raw skill the two systems are equally reliable; their score-spread curves cross at a 49% strike rate, but their discrimination does not.** A variance decomposition shows nearly identical single-game reliability across the entire skill range, and the larger spread traditional scoring carries at elite levels (over 30% more at the Top 10 tier) is mostly within-player noise. The crossover marks a change in the granularity of the score scale, not a reversal in how reliably a single game ranks two players. The case for traditional scoring does not rest on score spread, and a common version of that case is mistaken.

4. **The genuine and exclusive advantage of traditional scoring is sequencing discrimination.** At a fixed strike rate, traditional scores rise with a player's tendency to string strikes together (population correlation 0.49; Cohen's $d = 0.23$ between the extremes), while World Bowling scores stay flat (correlation -0.04). World Bowling is not merely a noisier measure of this dimension; by commutativity (Proposition 1) it is structurally blind to it. The per-game effect is moderate but compounds: as an upper bound, a maximally streaky player versus a streak-free player of identical raw skill wins a head-to-head 48-game tournament about 90% of the time under traditional scoring, against roughly chance (53%) under World Bowling, with the true figure depending on the dispersion of streakiness in the population, which is not yet established. The advantage is specific to strikes; clustering spare conversions carries no signal under either system. This is the property the choice between the two systems actually turns on.

The complexity of traditional scoring is not an accident of history. It is the mechanism by which the system encodes one quality distinctive of elite bowling: the ability to string consecutive strikes across frames and across games. Removing that complexity removes that measurement, while leaving the measurement of raw skill essentially intact.

For governing bodies, tournament organisers, and league administrators, the practical implication is a clarified trade-off rather than a verdict. Current-frame scoring loses little in the measurement of raw skill and gains real simplicity, which makes it well suited to recreational and casual play. Its cost falls specifically on sequencing: it cannot distinguish a player who sustains consecutive strikes from one who scatters the same strikes through a game. Whether elite competition should reward that sequencing is a normative decision about what the sport wishes to measure; this paper's contribution is to show precisely what is kept and what is discarded, so that the decision can be made deliberately rather than by appeal to score spread.

Reproducibility

All results in this paper are fully reproducible. The source code, exact distribution data, and analysis scripts are publicly available at https://github.com/michael-borck/skill-sequence-scoring. An interactive companion website at https://michael-borck.github.io/skill-sequence-scoring allows readers to explore the sequence sensitivity and reward gradient results directly.

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